

B. Tech Degree VI Semester (Supplementary) Examination September 2010

ME 606 CAD/CAM
(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART – A
(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Describe any three levels of automation.
 (b) Explain Detroit-type automation.
 (c) What is numerical control?
 (d) Differentiate between open loop and closed loop systems.
 (e) What are the advantages of CNC controlled production machine tools?
 (f) Discuss the various elements of information processing unit in NC machine tool.
 (g) Write short note on sensors in industrial robots.
 (h) Explain the applications of industrial robots.

PART – B

(4 x 15 = 60)

- II. (a) Explain geometric modelling in computer aided design. (7)
 (b) Explain the data exchange between CAD and CAM. (8)

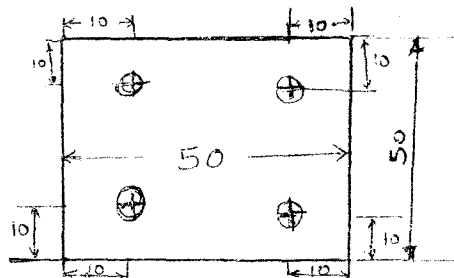
OR

- III. (a) Explain any two methods of line balancing. (10)
 (b) Describe any five ways to improve line balancing. (5)

- IV. (a) What is CNC? What are the control features of CNC? (10)
 (b) Write a note on preparatory functions. (5)

OR

- V. (a) It is required to drill four holes, centered at the four corners of a rectangular formation as shown in the figure. Write an APT program. Depth of hole = 10 mm



Spindle (-10,0,20)

- (b) Describe any three types of statements in APT language. (12)

- VI. With a neat diagram, explain the features of a CNC system for accurate and efficient cutting. (15)

OR

- VII. Describe - (i) CNC systems for horizontal machining centre (7)
 (ii) CNC systems for lathe. (8)

- VIII. (a) With sketches, explain any five basic structures of industrial robots. (10)
 (b) Discuss the classification of robots. (5)

OR

- IX. Explain -
 (i) Direct Numerical Control
 (ii) Computer Integrated Manufacture
 (iii) Flexible Manufacturing System

(15)

